MCQS SOLVED BY ***HASNAIN AZEEZ***

1. l-form bacteria lack (**cell wall**)
2. glutamic acid producing genra except (**Saccharomyces**)
3. 3 steps of Calvin cycle include (**regeneration**)
4. Microbial enzymes are predominently produced by(**Submerged fermentation**)
5. Gasohol, a b;end of _____(**10-22%**) v/v ethanol with gasoline
6. Insoluble Inactive aggregates of overexpressed polypetides_____ (**Inclusion bodies**)
7. _____ used to control pH (**ammonium Hydroxide**)
8. ___ are used to control anabolic reactions (**repression**
9. **Bioinformatics** is the use of computing technologies to manage and analyse biological data.
10. Fully developed spherical rod shaped organelles enclosed by double membrane areb called **mitochondria**
11. During **batch** fermentation the population of microorganisms goes through several distinct growth phases of log, lag, acceleration, stationar n death,
12. **Conjugation** involves cell-to-cell contact, where the donor contacts the recipient with a filamentous protein structure called a sex pilus
13. Restriction enzyme cut double stranded DNA at specific sequence called **palindrome** ,
14. Whey is an aqueous byproduct of **dairy** industry,
15. Manton and Gaulin homogenizer (APV-type mill), are employed for pilot- and production-scale **cell disruption.**
16. Term beer is given to **non-distilled alcoholic** beverages made from cereal grains,
17. Primary metabolites are **active** growth phase or **trophophase**,
18. Alcoholic beverages are produced in (_**non aseptic**_) conditions,
19. Advantage of solid substrate fermentation except (_**slow microbial growth**_),
20. Gasohol is blend of **10–22%** v/v combination of gasoline n ethanol
21. Heat is generated during fermentation process primarily do to metabolic activity and agitation process
22. In airlift fermenter Liquid movement is initiated by the injection of **Compressed air** at the bottom of the internal or external Riser column
23. In spread plating, samples volume is usually **0.1ml**

24. **White Biotechnology also known as industrial biotechnology** applies to industrial processes. Designing of an organism to use to produce a useful chemical.

Subjective solved by ZAREEN FATIMA

citric acid uses in industries 2

• Citric acid is widely used in the food industry as an acidulant and flavouring agent in beverages, confectionery and other foods, and in leavening systems for baked goods.

• As a food constituent, its use is unrestricted because it has GRAS status.

• They include roles in maintaining metals in solution for electroplating, as a cleaning and 'pickling' agent for metals, and as a replacement for polyphosphates in the detergent industry, along with several pharmaceutical uses.

factors influencing oxygen transfer 2

• Oxygen transfer is complex, as it involves a phase change from its gaseous phase to the liquid phase, and is influenced by the following factors.

1. The prevailing physical conditions; temperature, pressure and surface area of air/oxygen bubbles;

2. The chemical composition of the medium;

3. The volume of gas introduced per unit reactor volume per unit time;

4. The type of sparger system used to introduce air into the fermenter;

5. The speed of agitation; or

6. A combination of these factors

Metabolic Regulation 2

• Microorganisms naturally have tight control over their metabolism,

• But this often needs to be overcome when they are used in industrial fermentations, e.g. In attempting to over-produce specific metabolites.

• Consequently, the study of metabolic regulation is of special importance in industrial microbiology.

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• REGULATORY ENZYMES

• Many metabolic pathways contain regulatory enzymes whose activity can be adjusted to provide a mechanism for fine tuning the flux of carbon through a pathway.

• Some of these enzymes are allosteric, usually composed of several subunits, and do not follow michaelis– menten kinetics.

Foam production and its disadvantage 3

• Foaming is largely due to media proteins that become attached to the air–broth interface where they denature to form a stable foam.

• If uncontrolled the foam may block air filters, resulting in the loss of aseptic conditions; the fermenter becomes contaminated and microorganisms are released into the environment.

Disadvantages:

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Name 5 GRAS microbes 3

Expression in GRAS-listed organisms provides obvious advantages. Prime candidates for this role are species within three genera of microorganisms, namely Bacillus, Aspergillus and Saccharomyces.

Organisms name: *Saccharomyces cerevisiae*, *Saccharomyces fragilis*, *Candida utilis*, *Aspergillus niger*, *Endothia parasitica*, *Bacillus cereus*, *Mucor pusillus Lindt* and *Mucor miehei* and *Aspergillus oryzae*

yeast as eukaryotic host for heterogeneous gene 5

A preferred alternative strategy for the expression of heterologous eukaryotic genes is often the employment of a suitable eukaryotic host

that will naturally perform any necessary post-translational protein modification and secretion.

• Initially, saccharomyces cerevisiae was a popular choice because it is safe and a vast amount of information has been accumulated about its genetics, physiology and performance in industrial fermentation processes.

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• Also, it has a relatively rapid growth rate and readily undergoes genetic manipulation,

• but unliken higher eukaryotes (animal and plant cells) this yeast is Easily and cheaply grown on an industrial scale. Several heterologous eukaryotic proteins have been successfully mass produced by S. Cerevisiae.However, product yields are relatively low at 1–5% of total protein and some proteins are retained within the periplasm. • Other yeasts may be better hosts, particularly the methylotrophs, pichia angusta (formerly hansenula polymorpha) and pichia pastoris, which have a number of advantages over S. Cerevisiae.

• They have strong inducible promoters and are capable of generating posttranslational protein modification similar to those performed by human cells. • Downstream processing is also less problematical as they do not secrete many of their own proteins into the medium.

Ways of cell harvesting 3

Broth conditioning Broth conditioning techniques are mostly used in association with sedimentation and centrifugation for the separation of cells from liquid media. Some organisms naturally flocculate, which can be enhanced by

chemical, physical and biological treatments. Such treatments can also be effective with cells that would not otherwise form flocs

Sedimentation Sedimentation is extensively used for primary yeast separation in the production of alcoholic beverages, and in waste-water treatment.

Centrifugation If instead of simply using gravitational force to separate suspended particles, a centrifugal field is applied, the rate of solid–liquid separation is significantly increased and much smaller particles can be separated.

STR 5

• Stirred tank reactors (STRs) have mechanically moving

• Agitators or impellers within a baffled cylindrical vessel.

• Baffles are usually flat vertical plates whose width is about one-tenth of the vessel diameter.

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• Normally, 4–6 baffle plates are fitted to the inside vessel walls to aid mixing and mass transfer by increasing turbulence, preventing vortex formation and eliminating ‘dead spaces’.

• Strs are the most commonly used vessels and have been adopted for a wide range of fermentation processes.

• Within each vessel the impeller is connected to an external motor which drives the stirrer system. The agitator assembly, including the seal, is often a potential route of contamination.

• To prevent this problem, the shaft has to pass into the fermenter through a set of aseptic seals.

• There are specific regulations regarding the numbers and types of seals.

• For certain fermentations, two or three seals are required to minimize the risk of fermenter contamination, and release of microorganisms and their products into the environment.

uses of microbial enzymes in industries 5

Commercial microbial enzymes are increasingly replacing conventional chemical catalysts in many industrial processes.

Certain enzymes are not restricted to aqueous environments and can operate in twophase water–organic solvent systems and in non-aqueous organic media, particularly hydrophobic solvents.

Several thousand tonnes of commercial enzymes are currently produced each year, which have a value in excess of US$1500 million.

Applications:

• Besides their role as aids in traditional processes, ‘bulk’ enzymes are at the centre of many novel processes and biotechnological innovation continues to expand the range of applications. • Proteases are not the only enzymes used in detergents; since the late 1980s, amylases and lipases have been available for incorporation, e.g. Lipolase from Novo Nordisk. • A Swiss chemist Jaag, who worked for the detergent company Gebrüder Schnyder, developed a new product containing a bacterial protease, which replaced the animal trypsin that had previously been incorporated into these detergent products. • Microbial enzymes are predominantly produced by submerged fermentations, although

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some solid-substrate fermentations are used, particularly for the production of extracellular fungal enzymes

processes involved in recovery of products 10

Recovery of products:

Following cell disruption, soluble proteins are usually separated from cell debris by centrifugation.

The solubility of the salt varies with temperature, so strict temperaturE control is required.

Liquid Liquid Extraction method:

Many alkaloids, antibiotics, steroids and vitamins are recovered by liquid–liquid extraction methods using organic solvents.

Chromatographic techniques

Chromatographic techniques are usually employed for higher-value products.

These methods, normally involving columns of chromatographic media (stationary phase), are used for desalting, concentration and purification of protein preparations.

The order and choice of technique will depend on the particular product, but the following chromatographic parameters should be considered: capacity, recovery andresolving power (selectivity).

Affinity chromatography

Affinity chromatography is a particularly powerful and highly selective purification technique

Ligands are covalently linked to the matrix material, e.g. agarose, via a spacer arm to avoid steric hindrance

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The loading capacity of affinity columns is large, as the volume of the sample is unimportant and high resolution can be attained

For industrial-scale operations, it is often necessary to sterilize the chromatographic media in order to comply with various

regulatory requirements.

Gel filtration chromatography:

Gel filtration chromatography essentially involves separation on the basis of molecular size (molecular sieving), although molecular shape can also influence separation performance.

The initial choice of stationary phase material is also a key factor, as some may interact with the target product, e.g. carbohydrate-based matrices interact with glycoprotein's.

High-performance liquid chromatography(HPLC)

High-performance liquid chromatography(HPLC) was originally developed for the separation of organic molecules in non-aqueous solvents, but is now used for proteins in aqueous olution.

The protein to be purified must have an affinity for this ion and binds to it by forming coordination complexes with groups such as the imidazole of histidine residues.

Membrane separation techniques:

membrane separation techniques are primarily used for the removal of low molecular weight solutes and inorganic ions from a solution

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